

$$D\nabla^4 w + \rho h \ddot{w} = f_d(t) \delta(x - x_d) \delta(y - y_d) - \sum_{i=1}^3 \delta(x - x_i) \delta(y - y_i) [k_i(w - u_i) + c_i(\dot{w} - \dot{u}_i)]$$

$$m_i \ddot{u}_i + c_i(\dot{u}_i - \dot{w}_i) + k_i(u_i - w_i) = 0, \quad w_i = w(x_i, y_i, t), \quad i = 1, 2, 3$$

$$D = \frac{Eh^3}{12(1 - \nu^2)}, \quad k_i = m_i (2\pi f_{t,i})^2$$